

WASP-10b

R-0702

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_231107 · created 2026-07-27T21:54:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460255.6897 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1582 +/- 0.0082
Transit depth (Rp/Rs)^2	2.5 +/- 0.26 [%]
Orbital Inclination (inc)	87.91 +/- 0.7
Airmass coefficient 1 (a1)	0.988 +/- 0.012
Airmass coefficient 2 (a2)	0.009 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	#1 - [167, 439]
Optimal Aperture	3.64
Optimal Annulus	17.31
Transit Duration (day)	0.0879 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1582 ± 0.0082 vs 0.1592 (deviation 0.09σ)
Duration fit vs published	0.0879 d vs 0.0946 d (deviation 0.97σ)
Mid-time O–C	-0.8 min
Depth SNR	14.6
Residual scatter	1.18 %

Light curve

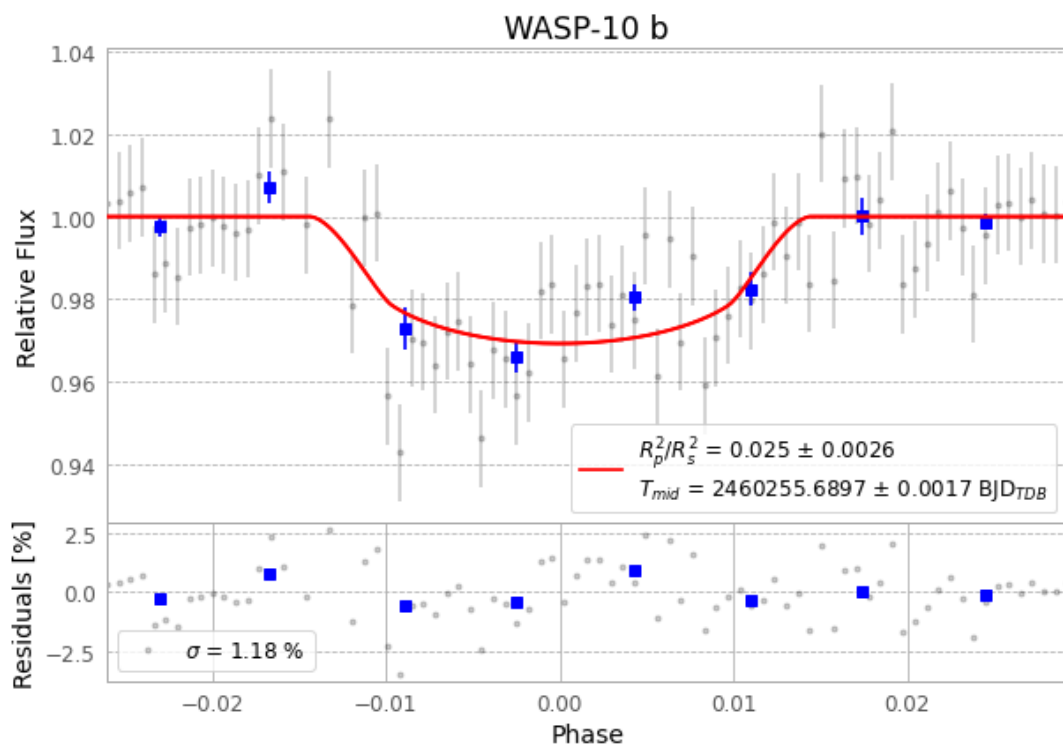


Figure 1 — FinalLightCurve_WASP-10 b_2023-11-06.png (SHA-256 64540f1abc03b298...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7c995c1bda0a1c33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

83 FITS frames (55 MB), WASP-10231107023015.FITS → WASP-10231107063615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231107.log (15d411d4497b...), FinalLightCurve_WASP-10 b_2023-11-06.png (64540f1abc03...), FinalLightCurve_WASP-10 b_2023-11-06.csv (cb694bb557f4...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:54Z.